

# Joins

*Introduction to QGIS*

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# Data Merging in GIS

Techniques for merging data from 2 separate layers in a GIS

1. Table Joins
2. Spatial Joins
3. Raster Sampling

# Table Joins

Join a vector layer with a table.

Need to have a field in both layers with a common identifier

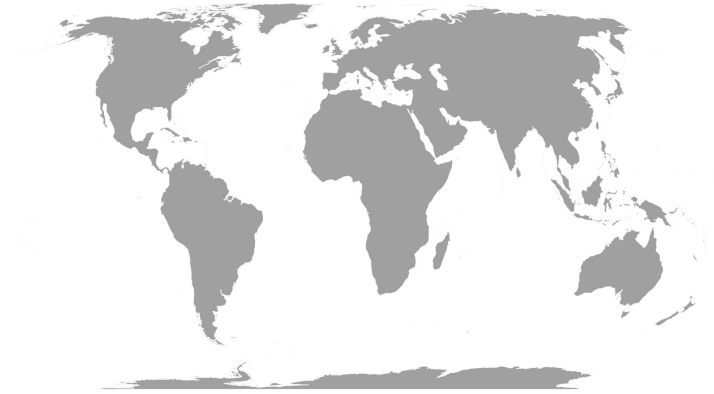
# Table Join Example

You have a shapefile of all countries with an attribute called “name” containing the name of the country

You also have an Excel file with 2 columns. A “country” column with country names and a “GDP per Capita” column.

You need to create a map of GDP per capita.

## Vector Layer

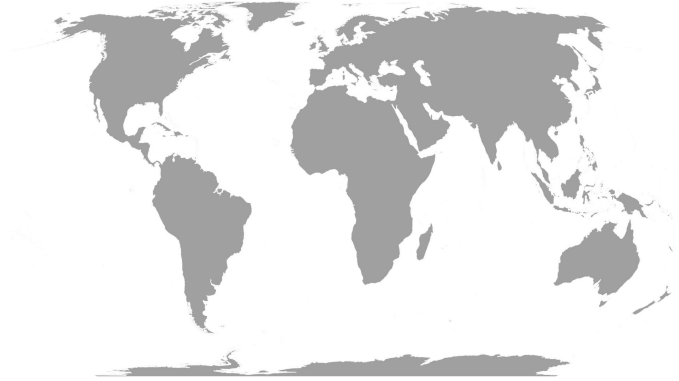


| <i><b>fid</b></i> | <i><b>name</b></i> | <i><b>code</b></i> |
|-------------------|--------------------|--------------------|
| 1                 | Afghanistan        | AF                 |
| 2                 | Albania            | AL                 |
| ..                | ..                 | ..                 |

## Table

| <i><b>country</b></i> | <i><b>GDP per capita (USD)</b></i> |
|-----------------------|------------------------------------|
| Afghanistan           | 508                                |
| Albania               | 5215                               |
| ..                    | ..                                 |

## New Vector Layer



| <i><b>fid</b></i> | <i><b>name</b></i> | <i><b>code</b></i> | <i><b>GDP per capita<br/>(USD)</b></i> |
|-------------------|--------------------|--------------------|----------------------------------------|
| 1                 | Afghanistan        | AF                 | 508                                    |
| 2                 | Albania            | AL                 | 5215                                   |
| ..                | ..                 | ..                 | ..                                     |

# Table Join in QGIS

Join Attributes by Field Value

# Spatial Joins

Join 2 vector layers based on their spatial relationships.

# Spatial Join Example 1

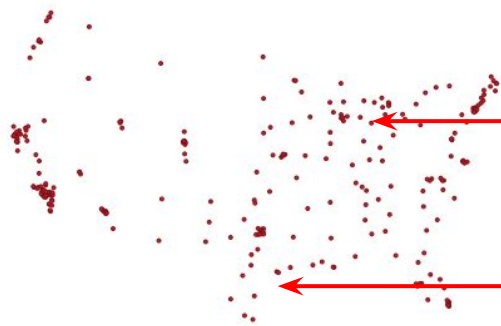
You have a Point layer of 'cities'.

You have a Polygon layer of 'states'.

You want to add the name of the state that each city belongs to.

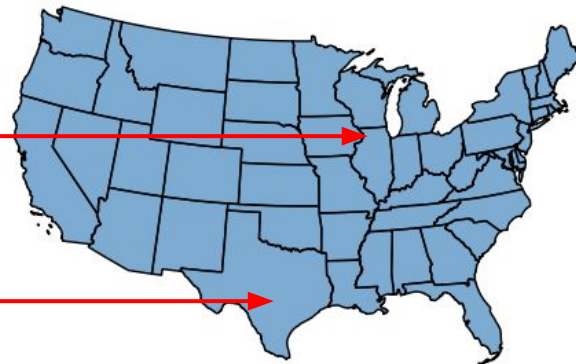
**Join Attributes by Location**

## Cities



| <i>fid</i> | <i>name</i> | <i>population</i> |
|------------|-------------|-------------------|
| 1          | Houston     | 2099451           |
| 2          | Chicago     | 2695598           |
| ..         | ..          | ..                |

## States



| <i>fid</i> | <i>name</i> | <i>code</i> |
|------------|-------------|-------------|
| 1          | Texas       | TX          |
| 2          | Illinois    | IL          |
| ..         | ..          | ..          |

## Cities with State Names



| <i>fid</i> | <i>name</i> | <i>population</i> | <i>state</i> |
|------------|-------------|-------------------|--------------|
| 1          | Houston     | 2099451           | Texas        |
| 2          | Chicago     | 2695598           | Illinois     |
| ..         | ..          | ..                |              |

# Spatial Join Example 2

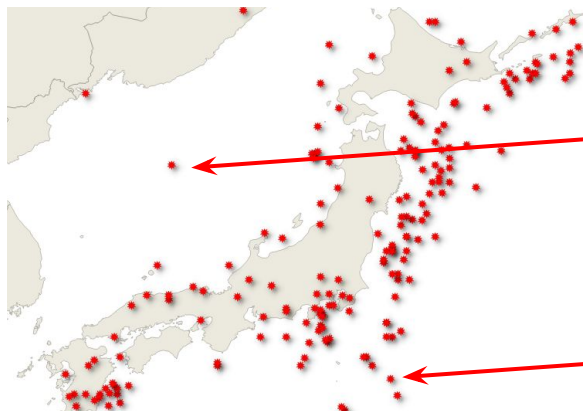
You have a Point layer of 'earthquakes'.

You have a Point layer of 'populated places'.

You want to find the nearest populated place for each earthquake.

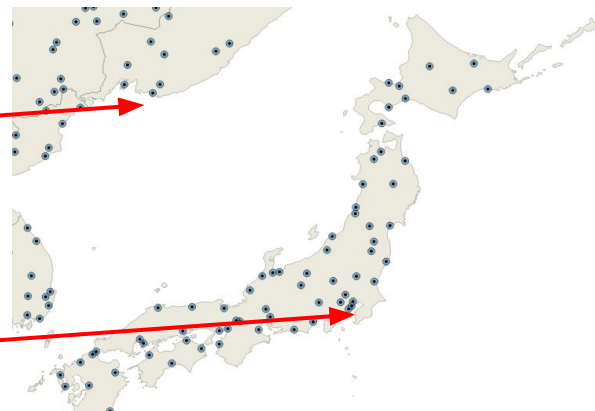
**Join Attributes by Nearest**

## Earthquakes



| <i>fid</i> | <i>date</i> | <i>magnitude</i> |
|------------|-------------|------------------|
| 1          | 1894-06-20  | 6.6              |
| 2          | 2018-06-18  | 5.5              |
| ..         | ..          | ..               |

## Populated Places



| <i>fid</i> | <i>name</i> | <i>population</i> |
|------------|-------------|-------------------|
| 1          | Tokyo       | 37393000          |
| 2          | Osaka       | 2753862           |
| ..         | ..          | ..                |

## Earthquake with Closest Populated Place



| <i>fid</i> | <i>date</i> | <i>magnitude</i> | <i>place</i> |
|------------|-------------|------------------|--------------|
| 1          | 1894-06-20  | 6.6              | Tokyo        |
| 2          | 2018-06-18  | 5.5              | Osaka        |
| ..         | ..          | ..               |              |

# Spatial Join Example 3

You have a Polygon layer of “districts”.

You have a Line layer of “roads”.

You want to calculate the length of roads in each district.

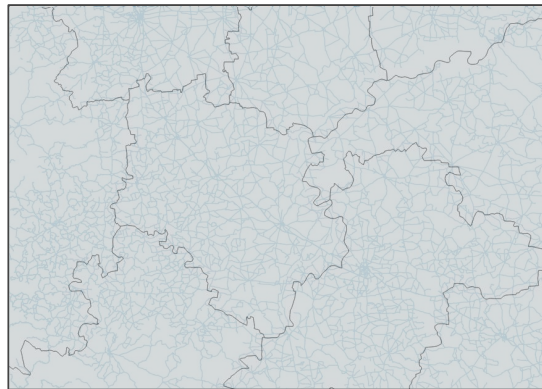
**Join Attributes by Location (Summary)**

## Districts



| <i><b>fid</b></i> | <i><b>name</b></i> | <i><b>code</b></i> |
|-------------------|--------------------|--------------------|
| 1                 | Bellary            | 1                  |
| 2                 | Belgaum            | 2                  |
| ..                | ..                 | ..                 |

## Roads



| <i><b>fid</b></i> | <i><b>segment_id</b></i> | <i><b>length_km</b></i> |
|-------------------|--------------------------|-------------------------|
| 1                 | 1234                     | 10.5                    |
| 2                 | 2345                     | 3.4                     |
| ..                | ..                       | ..                      |

## District with Lengths



| <i><b>fid</b></i> | <i><b>name</b></i> | <i><b>code</b></i> | <i><b>length_sum</b></i> |
|-------------------|--------------------|--------------------|--------------------------|
| 1                 | Bellary            | 1                  | 1356.4                   |
| 2                 | Belgaum            | 2                  | 756.3                    |
| ..                | ..                 | ..                 |                          |

# Spatial Joins in QGIS

Join Attributes by Location

Join Attributes by Nearest

Join Attributes by Location (Summary)

# Raster Sampling

Join a vector layer with a raster layer

# Raster Sampling Example 1

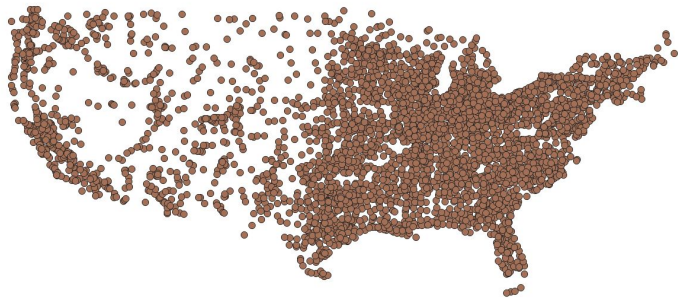
You have a point layer of 'cities'.

You have a Raster layer of 'Maximum Temperature'.

You need the maximum temperature at each city.

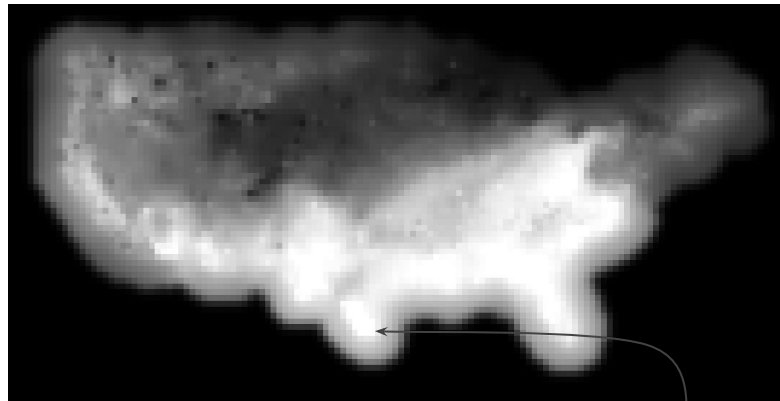
**Sample Raster Values**

## Cities



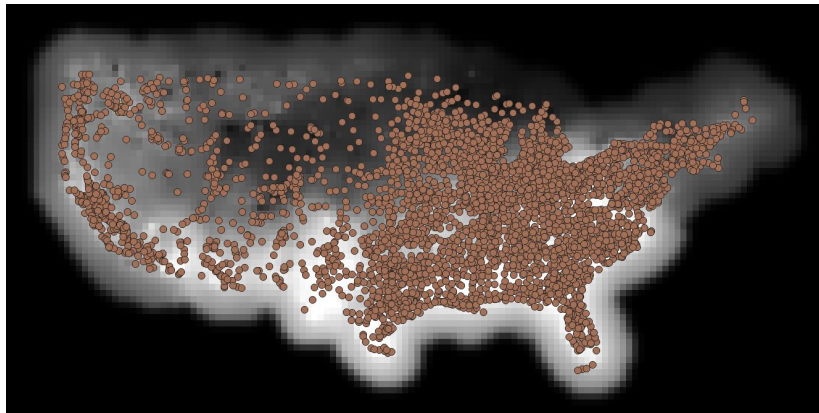
| <i><b>fid</b></i> | <i><b>name</b></i> | <i><b>state</b></i> |
|-------------------|--------------------|---------------------|
| 1                 | Albany             | NY                  |
| 2                 | El Paso            | TX                  |
| ..                | ..                 | ..                  |

## Temperature (°C)



31.7

## Cities with Temperature



| <i><b>fid</b></i> | <i><b>city</b></i> | <i><b>state</b></i> | <i><b>temperature</b></i> |
|-------------------|--------------------|---------------------|---------------------------|
| 1                 | Albany             | NY                  | 22.3                      |
| 2                 | El Paso            | TX                  | 31.7                      |
| ..                | ..                 | ..                  |                           |

# Raster Sampling Example 2

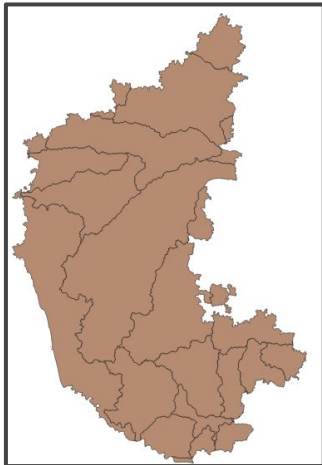
You have a Polygon layer of “river basins”.

You have a Raster layer of “rainfall”.

You want to calculate the average rainfall in each basin

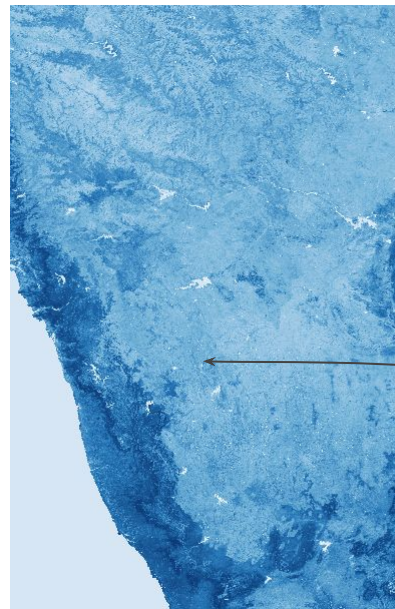
**Zonal statistics**

## River Basins



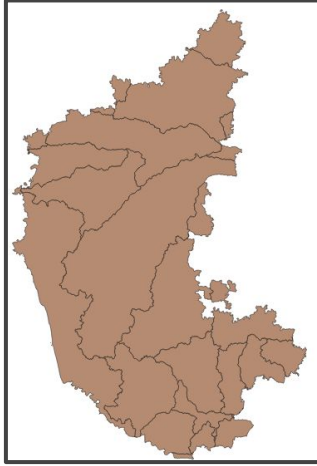
| <i><b>fid</b></i> | <i><b>basin_name</b></i> | <i><b>basin_id</b></i> |
|-------------------|--------------------------|------------------------|
| 1                 | Cauvery                  | C1                     |
| 2                 | Krishna                  | K9                     |
| ..                | ..                       | ..                     |

## Gridded Rainfall 2020 (mm/year)



787

## River Basins with Rainfall



| <i><b>fid</b></i> | <i><b>basin_name</b></i> | <i><b>basin_id</b></i> | <i><b>average_rainfall</b></i> |
|-------------------|--------------------------|------------------------|--------------------------------|
| 1                 | Cauvery                  | C1                     | 1156                           |
| 2                 | Krishna                  | K9                     | 952.5                          |
| ..                | ..                       | ..                     |                                |

# Raster Sampling in QGIS

Sample Raster Values

Zonal statistics

# Raster Sampling Example 3

You have a Line layer of 'streets'.

You have a Raster layer of 'Elevation'.

You need to know the gradient of each street.

Drape (Set Z value from Raster))